Python uv Cheatsheet

The Fast, Modern Python Package Manager

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First edition

This book was professionally typeset on Reedsy.
Find out more at reedsy.com

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What is uv?

- A super-fast, modern Python package and environment manager by **Astral**
- Written in **Rust**, compatible with pip, venv, and poetry workflows
- Drop-in replacement for pip + venv + pip-tools combo

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Installation

Install via script:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Or with Homebrew:

```
brew install uv
```

Verify:

```
uv --version
```

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Creating & Managing Environments

Create a virtual environment:

```
uv venv
```

Activate it:

```
source .venv/bin/activate   # macOS/Linux
.venv\Scripts\activate      # Windows
```

Remove environment:

```
uv venv --remove
```

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Installing & Managing Packages

Install a package:

```
uv add requests
```

Install multiple:

```
uv add flask numpy pandas
```

Remove package:

```
uv remove flask
```

Sync all dependencies:

```
uv sync
```

Upgrade packages:

```
uv update
```

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Requirements & Lock Files

Generate lock file:

```
uv lock
```

Export to requirements.txt:

```
uv export -o requirements.txt
```

Install from lock file:

```
uv sync
```

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Running Scripts

Run a Python file inside the environment:

```
uv run main.py
```

Or run any command:

```
uv run pytest
uv run python manage.py runserver
```

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Project Initialization

Start a new project:

```
uv init myproject
cd myproject
```

Auto-detect dependencies:

```
uv add -r requirements.txt
```

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Comparing uv vs pip / poetry

Task	pip + venv	poetry	uv
Create env	`python -m venv`	auto	`uv venv`
Install pkg	`pip install`	`poetry add`	`uv add`
Lock deps	`pip-tools`	auto	`uv lock`
Speed	🐢	🐇	⚡ Rust-fast

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Useful Flags & Tricks

Install from Git:

```
uv add git+https://github.com/user/repo.git
```

Install in editable mode:

```
uv add -e .
```

Reinstall clean:

```
uv sync --clean
```

Use specific Python version:

```
uv python install 3.12
uv python list
```

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Example Workflow

```
# 1. Init new project
uv init myapp
cd myapp

# 2. Add dependencies
uv add fastapi uvicorn

# 3. Run project
uv run uvicorn main:app --reload
```

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Bonus: Integration Tips

- Works seamlessly with pyproject.toml
- Compatible with GitHub Actions & Docker
- Can replace requirements.txt entirely

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Cleanup & Maintenance

Remove unused packages:

```
uv prune
```

Clear cache:

```
uv cache clean
```

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Resources

Official Docs

GitHub Repository

Astral Blog